

Single-cell analysis assistant

Upload a dataset, ask questions, and browse generated plots in a scientific chat interface with guarded interpretation and reproducible outputs.

Dataset setup

Hide

Generic roles on the left, your dataset's columns on the right. Confirm or change them — every analysis follows this table.

WHAT IT IS	COLUMN IN YOUR DATA	DETAILS
Cell type	cell_ontology_class — suggest ▼	6 type(s)
Sample replicate unit	sample_id — suggested ▼	13 samples
Compare by grouping variable	comparison_group — suggest ▼	group_1 (4), group_2 (6)
Adjust for DESeq2 covariates	☐ FACS.selection ☒ sex	

Apply Saved ✓



Data preview

tabula-muris-senis-facs-processed-official-annotations-Spleen.h5ad

3,834 cells x 22,966 genes ▼

cell_id	FACS.selection	age	cell	cell_ontology_class
A10_B002452_B009020_S10.mm10-plus-0...	Viable	18m	A10_B002452	B cell
A11_B002452_B009020_S11.mm10-plus-0...	Viable	18m	A11_B002452	B cell
A11_D045342_B009019_S11.mm10-plus-0...	Viable	18m	A11_D045342	B cell
A12_B002452_B009020_S12.mm10-plus-0...	Viable	18m	A12_B002452	B cell

cell_id	FACS.selection	age	cell	cell_ontology_class
A12_D045342_B009019_S12.mm10-plus-0...	Viable	18m	A12_D045342	CD4-positive, alpha-beta T cell
A13_B002452_B009020_S13.mm10-plus-0...	Viable	18m	A13_B002452	CD8-positive, alpha-beta T cell
A13_D045342_B009019_S13.mm10-plus-0...	Viable	18m	A13_D045342	B cell
A14_B002452_B009020_S14.mm10-plus-0...	Viable	18m	A14_B002452	CD8-positive, alpha-beta T cell
A15_B002452_B009020_S15.mm10-plus-0...	Viable	18m	A15_B002452	CD4-positive, alpha-beta T cell
A15_D045342_B009019_S15.mm10-plus-0...	Viable	18m	A15_D045342	B cell
A16_B002452_B009020_S16.mm10-plus-0...	Viable	18m	A16_B002452	B cell
A16_D045342_B009019_S16.mm10-plus-0...	Viable	18m	A16_D045342	CD4-positive, alpha-beta T cell
A17_B002452_B009020_S17.mm10-plus-0...	Viable	18m	A17_B002452	B cell
A18_B002452_B009020_S18.mm10-plus-0...	Viable	18m	A18_B002452	B cell
A18_D045342_B009019_S18.mm10-plus-0...	Viable	18m	A18_D045342	B cell
A19_B002452_B009020_S19.mm10-plus-0...	Viable	18m	A19_B002452	CD8-positive, alpha-beta T cell
A19_D045342_B009019_S19.mm10-plus-0...	Viable	18m	A19_D045342	B cell
A1_B002452_B009020_S1.mm10-plus-0-0	Viable	18m	A1_B002452	CD8-positive, alpha-beta T cell
A1_D045342_B009019_S1.mm10-plus-0-0	Viable	18m	A1_D045342	B cell
A20_B002452_B009020_S20.mm10-plus-0...	Viable	18m	A20_B002452	B cell



Showing 20
of 3,834
observation
rows

Genes: 0610005C13Rik, 0610007C21Rik, 0610007L01Rik, 0610007N19Rik, 0610007P08Rik, 0610007P14Rik, 0...

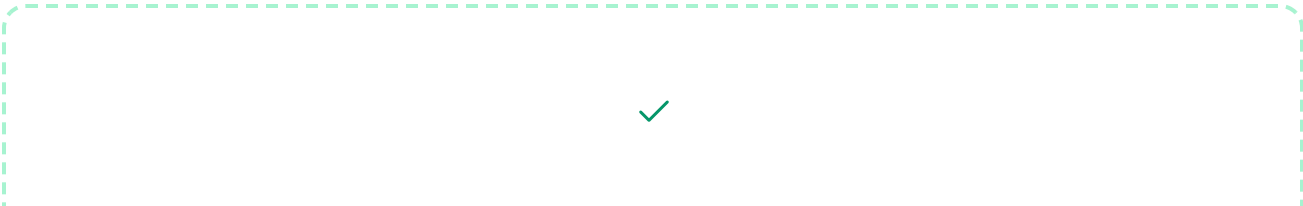
Dataset

Upload a .h5ad file to begin analysis.

Species (gene symbols)

Mouse

Species locked for this upload. Reset session to change.



Dataset loaded

tabula-muris-senis-facs-process...

[Click to replace](#)

READING RESULTS

Replies are governed: claims are limited to what the statistics support.

LOW POWER

Too few replicates/cells for a reliable test. Numeric trends only.

DESCRIPTIVE ONLY

Medians and counts reported. No p-values or causal claims.

NOT SIGNIFICANT

Test ran; difference not significant at $\alpha = 0.05$.

Senescence Agent

tabula-muris-senis-facs-processed-official-annotations-Spleen.h5ad

Report (1)



SKIV212

-2.08

2.141e-06

Positive log2FC = higher expression in 24m group. Pseudobulk aggregation across samples.

[Download all results \(CSV\)](#)

Caution (contrast admissible but imperfect):

the contrast is also explicitly encoded by ['comparison_group']. These registered aliases/sources were not treated as nuisance confounders and must not be added as covariates.

[Copy](#) > [Audit trail \(1\)](#)

TOOLS: [run deseq2](#)

Executed analysis

SIGNIFICANT INFERENTIAL

CONTRAST
24m vs 3m

GROUPING COLUMN
age

BIOLOGICAL UNIT
sample_id

COVARIATES USED
sex



Planner

accepted



Counts layers[counts]



Design sex + age

PLAUSIBILITY

ok

DONOR STABILITY

1995 / 2246 (0.8882)

SAMPLES

10

DROPPED COVARIATES

None

QUICK

SenMayo coverage

Score cells

T cell p-value

Age comparison

UMAP

DESeq2

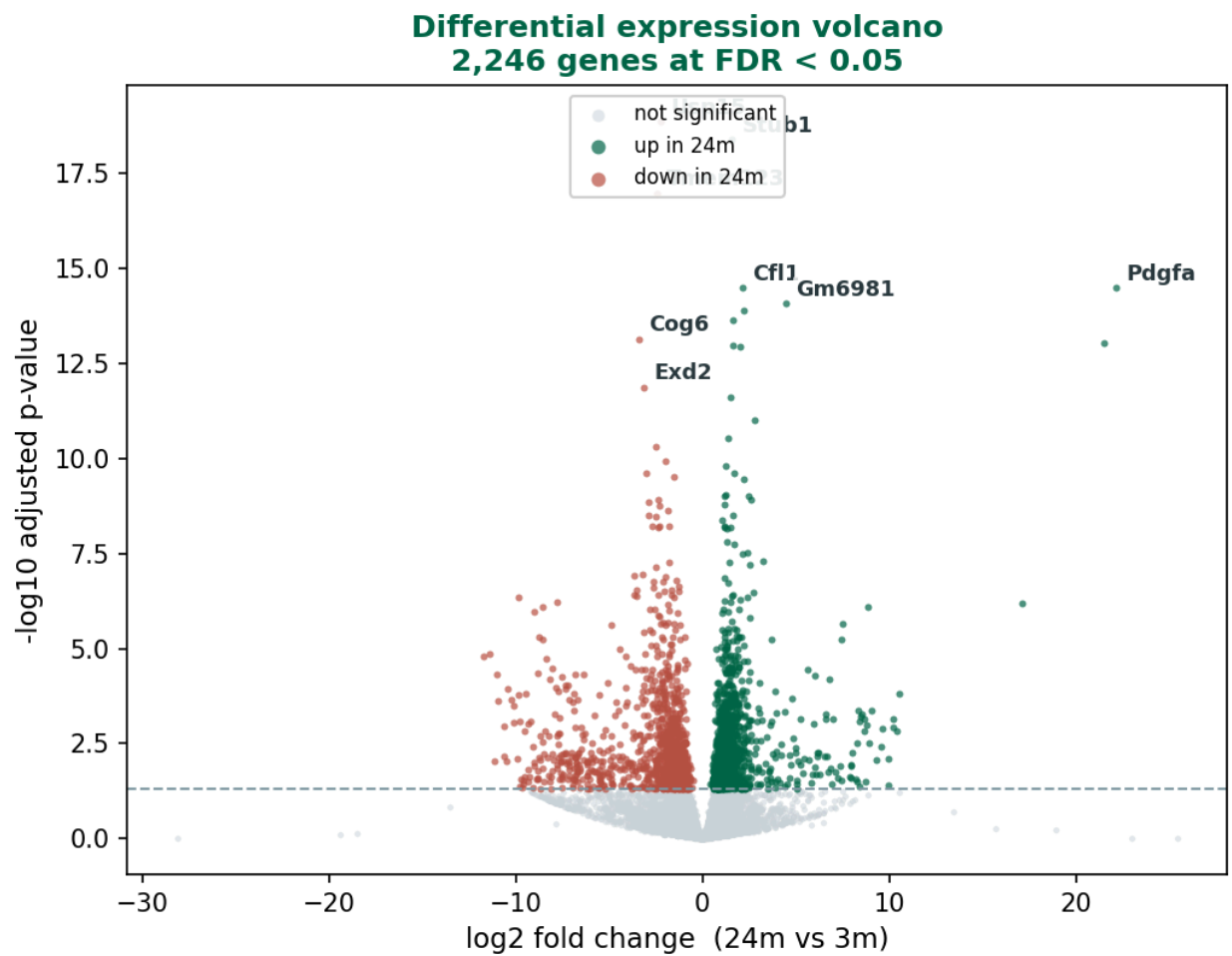
Ask about your dataset...

Enter send · Shift+Enter new line



Generated plots

1 plot in this session

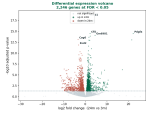


Run Deseq2

Generated analysis plot.

/plots/volcano.png

PLOT LIBRARY



Run Deseq2

Generated analysis plot.

